Prior Sensitivity Analysis

2026-04-24

Setup

```r
library(UsingR)
```

```
## Warning: package 'UsingR' was built under R version 4.4.3

## Loading required package: MASS

## Loading required package: HistData

## Warning: package 'HistData' was built under R version 4.4.3

## Loading required package: Hmisc

## Warning: package 'Hmisc' was built under R version 4.4.3

##
## Attaching package: 'Hmisc'

## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```r
library(Bolstad)
```

```
##
## Attaching package: 'Bolstad'

## The following objects are masked from 'package:stats':
##
##     IQR, sd, var
```

```r
babies_df <- babies
```

```r
# Remove all observations with missing or unknown values
babies_df = subset(babies_df, wt != 999 & parity != 99 & race != 99 & age != 99 &
                      ed != 9 & ht != 99 & wt1 != 999 & drace != 99 & dage != 99 &
                      ded != 9 & dht != 99 & dwt != 999 & inc != 99 & smoke != 9 &
                      time != 99 & time != 98 & number != 99 & number != 98)
```

Model

We fit a multiple linear regression of birth weight (oz) on six maternal predictors:

- gestation (days): longer pregnancies produce heavier babies
- ht (inches): taller mothers tend to have larger babies
- wt1 (lbs): heavier pre-pregnancy mothers tend to have heavier babies
- number: ordinal measure of cigarettes smoked per day (0=never, . . . , 8=60+)
- parity: 0 = first birth, 1 = previous birth(s)
- age (years): maternal age at delivery

```
wt ~ gestation + ht + wt1 + number + parity + age
```

There are 7 coefficients total (including intercept).

Prior Specifications

We compare four normal prior specifications. The prior mean b0 encodes our expectation about the direction and rough magnitude of each effect before seeing the data. The prior variance V0 encodes our uncertainty about those expectations. We use three prior mean vectors:

Prior 1 – Diffuse, zero mean (baseline) No prior knowledge assumed. Mean of zero on all coefficients, very large variance. This is the same prior structure used in the smoking-only model.

Prior 2 – Diffuse, zero mean, moderate variance Same zero-mean assumption but variance scaled down to 10 * I. This provides mild regularization without encoding any directional beliefs.

Prior 3 – Informative, research-backed means, diffuse variance Prior means are derived from published literature on each predictor's relationship with birth weight. Variance remains diffuse so the data can still move the posterior substantially. This prior asks: if we encode what existing research suggests about the direction and rough magnitude of each effect, do our posteriors change?

Prior mean derivations from literature review: - Intercept: set to 0 (absorbed into diffuse variance) - gestation: Oken et al. (2003) report median birth weight increments of ~140g per gestational week from a US national reference sample, independent of the CHDS. Converting to oz/day: 140g / 28.35g/oz / 7 days ~= 0.70 oz/day. We use 0.7. Source: Oken et al. (2003). BMC Pediatrics, 3:6. doi:10.1186/1471-2431-3-6 - ht: consistent positive association across studies; a 5cm increase in maternal height is associated with increases in newborn size (Polzlberger et al., 2016; Thame et al., 2004). We encode a weak positive prior of 1.0 oz/inch. - wt1: positive association well-established (Polzlberger et al., 2016). We encode a small positive prior of 0.05 oz/lb. - number: OLS estimates from multiple studies find each additional cigarette reduces birth weight by 8-15g (Juarez & Merlo, 2013). The `number` variable is ordinal with each step representing ~5 more cigarettes. Converting: 10g x 5 / 28.35 ~= -1.76 oz per category step. We use -1.5. Source: Juarez S, Merlo J (2013); Perreira et al. (2017) - parity: first births are on average 115-138g lighter than subsequent births (Wilcox et al., 1996; Bohn et al., 2021). Converting: 115g / 28.35 ~= 4 oz. Since parity = 1 means prior birth (i.e. NOT first birth), the coefficient should be positive. We use 4.0. Source: Bohn et al. (2021); Magnus et al. (1994) - age: modest positive effect up to early 30s (Stein et al., 1989). We encode a small positive prior of 0.1 oz/year. Source: Stein Z et al. (1989)

Prior 4 – Informative means, tighter variance on weaker predictors Same research-backed means as Prior 3, but variance is tightened on all of them (variance = 10 rather than 1e4).

```r
# --- Prior means ---

# Prior 1 & 2: zero mean (no prior belief)
b0_zero = rep(0, 7)
```

```r
# Prior 3 & 4: research-backed means
# Order: intercept, gestation, ht, wt1, number, parity, age
b0_informed = c(0,     # intercept: no prior
                0.7,   # gestation: ~0.7 oz/day (143g/week converted)
                1.0,   # ht: positive, ~1 oz/inch
                0.05,  # wt1: small positive, ~0.05 oz/lb
                -1.5,  # number: negative, ~-1.5 oz per category step
                4.0,   # parity: positive, ~4 oz for non-first birth
                0.1)   # age: small positive, ~0.1 oz/year

# --- Prior variances ---

# Prior 1: Diffuse (baseline)
V0_diffuse = 1e4 * diag(rep(1, 7))

# Prior 2: Moderate zero-mean
V0_moderate = diag(c(10, 10, 10, 10, 10, 10, 10))

# Prior 3: Diffuse variance, informed means
V0_diffuse_informed = 1e4 * diag(rep(1, 7))

# Prior 4: Informed means, tighter variance on all of them
# Order: intercept, gestation, ht, wt1, number, parity, age
V0_tight_weak = diag(c(10,   # intercept: diffuse
                       10,   # gestation: diffuse (strong signal)
                       10,   # ht: diffuse
                       10,   # wt1: diffuse
                       10,   # number: diffuse (strong signal)
                       10,     # parity: tighter (effect well-bounded ~4 oz)
                       10))    # age: tighter (effect small and well-bounded)
```

```r
fit1 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_zero,     V0 = V0_diffuse))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```r
fit2 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_zero,     V0 = V0_moderate))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```r
fit3 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_informed, V0 = V0_diffuse_informed))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```r
fit4 = bayes.lm(wt ~ gestation + ht + wt1 + number + parity + age,
                data   = babies_df,
                center = FALSE,
                prior  = list(b0 = b0_informed, V0 = V0_tight_weak))
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```r
# bayes.lm strips names from coefficients via as.vector(); restore them
coef_names = c("(Intercept)", "gestation", "ht", "wt1", "number", "parity", "age")
names(fit1$coefficients) = coef_names
names(fit2$coefficients) = coef_names
names(fit3$coefficients) = coef_names
names(fit4$coefficients) = coef_names
```

Posterior Summaries

```r
cat("=== Prior 1: Diffuse zero-mean (baseline) ===\n")
```

```
## === Prior 1: Diffuse zero-mean (baseline) ===
```

```r
summary(fit1)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##      age, data = babies_df, center = FALSE, prior = list(b0 = b0_zero,
##      V0 = V0_diffuse))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -68.296 -10.715  -0.009  10.883  53.440
##
## Coefficients:
##              Posterior Mean  Std. Error  t value
## (Intercept)   1.933e+01       3.049e+02   6.339e-02
## gestation     1.210e-02       7.369e-05   1.642e+02
## ht            1.400e+00       8.422e-02   1.662e+01
## wt1           6.079e-02       1.332e-03   4.562e+01
## number       -1.518e+00       9.758e-02  -1.556e+01
## parity       -2.851e-02       1.678e-01  -1.699e-01
## age           6.635e-02       1.862e-02   3.564e+00
## ---
## Prior Coefficients:
## (Intercept)    gestation           ht          wt1       number       parity
##           0            0            0            0            0            0
```

```
##         age
##           0
##
## Prior Covariance Matrix:
##              (Intercept)  gestation  ht     wt1    number  parity  age
## (Intercept)  10000            0          0      0      0       0       0
## gestation        0        10000          0      0      0       0       0
## ht               0            0      10000      0      0       0       0
## wt1              0            0          0  10000      0       0       0
## number           0            0          0      0  10000       0       0
## parity           0            0          0      0      0   10000       0
## age              0            0          0      0      0       0   10000
```

```r
cat("\n=== Prior 2: Moderate zero-mean ===\n")
```

```
##
## === Prior 2: Moderate zero-mean ===
```

```r
summary(fit2)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##      age, data = babies_df, center = FALSE, prior = list(b0 = b0_zero,
##      V0 = V0_moderate))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -68.468 -10.317   0.104  10.878  54.576
##
## Coefficients:
##              Posterior Mean  Std. Error  t value
## (Intercept)   6.417e-01       9.689e+00   6.623e-02
## gestation     1.303e-02       7.305e-05   1.783e+02
## ht            1.693e+00       1.069e-02   1.584e+02
## wt1           5.236e-02       1.265e-03   4.141e+01
## number       -1.521e+00       9.636e-02  -1.579e+01
## parity       -4.085e-02       1.649e-01  -2.478e-01
## age           9.345e-02       1.792e-02   5.216e+00
## ---
## Prior Coefficients:
## (Intercept)    gestation           ht          wt1       number       parity
##           0            0            0            0            0            0
##         age
##           0
##
## Prior Covariance Matrix:
##              (Intercept)  gestation  ht  wt1  number  parity  age
## (Intercept)  10            0          0   0    0       0       0
## gestation     0           10          0   0    0       0       0
## ht            0            0         10   0    0       0       0
## wt1           0            0          0  10    0       0       0
## number        0            0          0   0   10       0       0
```

```
## parity        0            0          0   0    0      10       0
## age           0            0          0   0    0       0      10
```

```r
cat("\n=== Prior 3: Informed means, diffuse variance ===\n")
```

```
##
## === Prior 3: Informed means, diffuse variance ===
```

```r
summary(fit3)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##      age, data = babies_df, center = FALSE, prior = list(b0 = b0_informed,
##      V0 = V0_diffuse_informed))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -68.296 -10.715  -0.009  10.883  53.440
##
## Coefficients:
##              Posterior Mean  Std. Error  t value
## (Intercept)   1.933e+01       3.049e+02   6.338e-02
## gestation     1.210e-02       7.369e-05   1.642e+02
## ht            1.400e+00       8.422e-02   1.662e+01
## wt1           6.079e-02       1.332e-03   4.562e+01
## number       -1.518e+00       9.758e-02  -1.556e+01
## parity       -2.845e-02       1.678e-01  -1.695e-01
## age           6.634e-02       1.862e-02   3.563e+00
## ---
## Prior Coefficients:
## (Intercept)    gestation           ht          wt1       number       parity
##        0.00         0.70         1.00         0.05        -1.50         4.00
##         age
##        0.10
##
## Prior Covariance Matrix:
##              (Intercept)  gestation  ht     wt1    number  parity  age
## (Intercept)  10000            0          0      0      0       0       0
## gestation        0        10000          0      0      0       0       0
## ht               0            0      10000      0      0       0       0
## wt1              0            0          0  10000      0       0       0
## number           0            0          0      0  10000       0       0
## parity           0            0          0      0      0   10000       0
## age              0            0          0      0      0       0   10000
```

```r
cat("\n=== Prior 4: Informed means, tight variance on age/parity ===\n")
```

```
##
## === Prior 4: Informed means, tight variance on age/parity ===
```

```r
summary(fit4)
```

```
##
## Call:
## bayes.lm(formula = wt ~ gestation + ht + wt1 + number + parity +
##      age, data = babies_df, center = FALSE, prior = list(b0 = b0_informed,
##      V0 = V0_tight_weak))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -69.001 -10.356  -0.026  10.799  54.597
##
## Coefficients:
##              Posterior Mean  Std. Error  t value
## (Intercept)   6.273e-01       9.689e+00   6.474e-02
## gestation     1.299e-02       7.305e-05   1.778e+02
## ht            1.699e+00       1.069e-02   1.589e+02
## wt1           5.126e-02       1.265e-03   4.054e+01
## number       -1.536e+00       9.636e-02  -1.594e+01
## parity        2.599e-02       1.649e-01   1.576e-01
## age           8.131e-02       1.792e-02   4.538e+00
## ---
## Prior Coefficients:
## (Intercept)    gestation           ht          wt1       number       parity
##        0.00         0.70         1.00         0.05        -1.50         4.00
##         age
##        0.10
##
## Prior Covariance Matrix:
##              (Intercept)  gestation  ht  wt1  number  parity  age
## (Intercept)  10            0          0   0    0       0       0
## gestation     0           10          0   0    0       0       0
## ht            0            0         10   0    0       0       0
## wt1           0            0          0  10    0       0       0
## number        0            0          0   0   10       0       0
## parity        0            0          0   0    0      10       0
## age           0            0          0   0    0       0      10
```

Comparison Table

```r
means1 = fit1$coefficients
means2 = fit2$coefficients
means3 = fit3$coefficients
means4 = fit4$coefficients

comparison = data.frame(
  Term          = names(means1),
  Prior_Mean    = round(b0_informed, 3),
  Post_Diffuse  = round(as.numeric(means1), 3),
  Post_Moderate = round(as.numeric(means2), 3),
  Post_Informed = round(as.numeric(means3), 3),
```

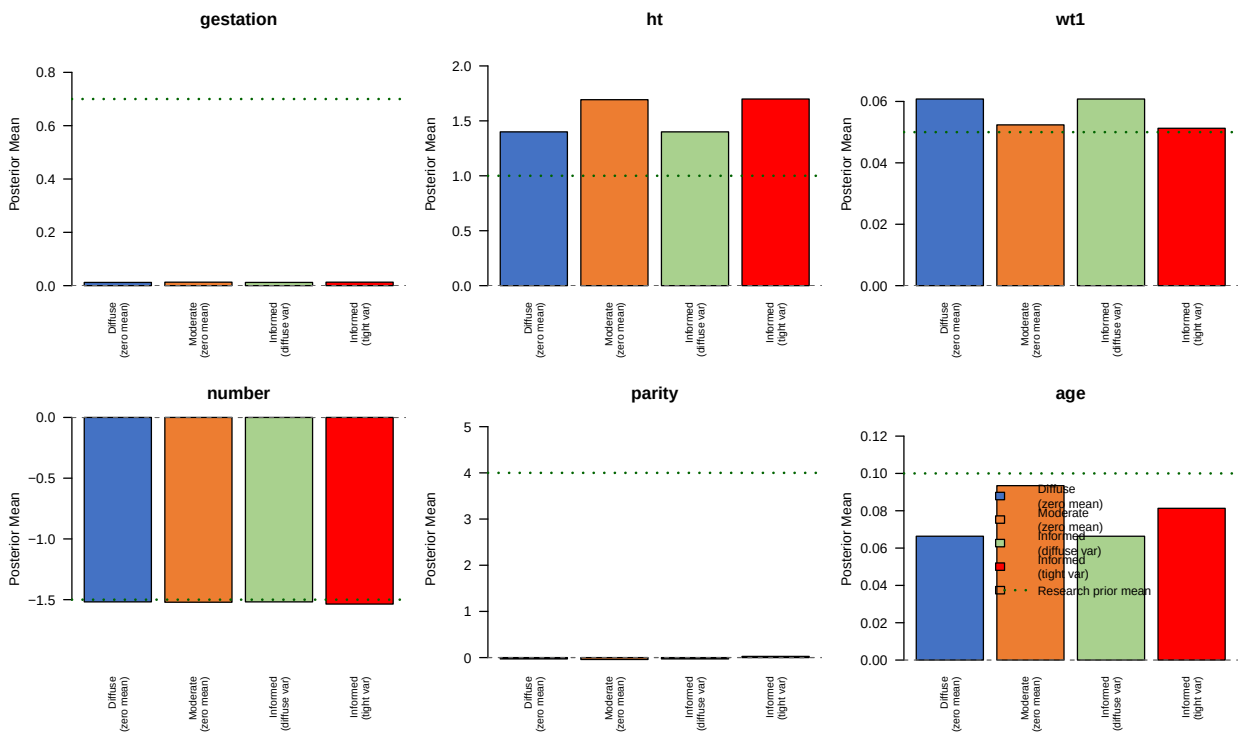
```r
  Post_Tight    = round(as.numeric(means4), 3)
)

# Add a column showing direction match with prior
comparison$Dir_Match = ifelse(
  sign(comparison$Prior_Mean) == 0 |
  sign(comparison$Post_Diffuse) == sign(comparison$Prior_Mean),
  "Yes", "No"
)

print(comparison)
```

```
##          Term Prior_Mean Post_Diffuse Post_Moderate Post_Informed Post_Tight
## 1 (Intercept)       0.00       19.326         0.642        19.326      0.627
## 2   gestation       0.70        0.012         0.013         0.012      0.013
## 3          ht       1.00        1.400         1.693         1.400      1.699
## 4         wt1       0.05        0.061         0.052         0.061      0.051
## 5      number      -1.50       -1.518        -1.521        -1.518     -1.536
## 6      parity       4.00       -0.029        -0.041        -0.028      0.026
## 7         age       0.10        0.066         0.093         0.066      0.081
##   Dir_Match
## 1       Yes
## 2       Yes
## 3       Yes
## 4       Yes
## 5       Yes
## 6        No
## 7       Yes
```

Visualization

```r
terms = names(means1)[-1]  # exclude intercept

prior_labels = c("Diffuse\n(zero mean)",
                 "Moderate\n(zero mean)",
                 "Informed\n(diffuse var)",
                 "Informed\n(tight var)")

cols = c("#4472C4", "#ED7D31", "#A9D18E", "#FF0000")

par(mfrow = c(2, 3), mar = c(5, 4, 3, 1))

for (term in terms) {
  idx = which(names(means1) == term)
  vals = c(means1[idx], means2[idx], means3[idx], means4[idx])
  prior_val = b0_informed[idx]

  bp = barplot(
    height    = vals,
    names.arg = prior_labels,
    main      = term,
```

```
    ylab      = "Posterior Mean",
    col       = cols,
    cex.names = 0.7,
    las       = 2,
    ylim      = range(c(vals, prior_val, 0)) * 1.3
  )
  abline(h = 0, lty = 2, col = "gray40")
  # Mark the research-based prior mean as a reference line
  abline(h = prior_val, lty = 3, col = "darkgreen", lwd = 2)
}


legend("center",
       legend = c(prior_labels, "Research prior mean"),
       fill   = c(cols, NA),
       lty    = c(NA, NA, NA, NA, 3),
       lwd    = c(NA, NA, NA, NA, 2),
       col    = c(rep("black", 4), "darkgreen"),
       bty    = "n",
       cex    = 0.9)
```



Credible Region Test

We test H0: all slope coefficients = 0 vs H1: at least one != 0. With 6 slope coefficients, the chi-square critical value at alpha=0.025 is 14.449.

```
run_cr_test = function(fit, label) {
  b1 = fit$coefficients[-1]
```

```r
  V1 = fit$post.var[-1, -1]
  cr = as.numeric(t(b1) %*% solve(V1) %*% b1)
  cat(label, "-> test statistic =", round(cr, 4))
  if (cr > 14.449) {
    cat("  (reject H0)\n")
  } else {
    cat("  (fail to reject H0)\n")
  }
}

run_cr_test(fit1, "Prior 1: Diffuse zero-mean     ")
```

```
## Prior 1: Diffuse zero-mean      -> test statistic = 66.0413  (reject H0)
```

```r
run_cr_test(fit2, "Prior 2: Moderate zero-mean    ")
```

```
## Prior 2: Moderate zero-mean     -> test statistic = 1427.548  (reject H0)
```

```r
run_cr_test(fit3, "Prior 3: Informed means, diffuse")
```

```
## Prior 3: Informed means, diffuse -> test statistic = 66.042  (reject H0)
```

```r
run_cr_test(fit4, "Prior 4: Informed means, tight ")
```

```
## Prior 4: Informed means, tight  -> test statistic = 1428.344  (reject H0)
```